

INTEGRATED ELECTRICAL SYSTEMS OF SOLAR PHOTOVOLTAIC TECHNOLOGY

LABORATORY 4

GROUP 2



TÉCNICO LISBOA

Experimental Investigation of the Effect of Tilt Angle on Photovoltaic Cell Performance

Authors:

Francisco Teixeira ist1102662
Manuel Barreiras ist1102554
Beatriz Coelho ist1107113

franciscorfteixeira@tecnico.ulisboa.pt
manuel.barreiras@tecnico.ulisboa.pt
beatriz.h.coelho@tecnico.ulisboa.pt

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1 Introduction

The primary objective of this laboratory experiment is to investigate the impact of the solar cell's tilt angle on its electrical performance. The tilt angle is directly related to the radiation's angle of incidence, which affects the effective irradiance that is absorbed by the cell. Therefore, optimizing the solar cell's angle is crucial to maximize the power output. To evaluate this effect, the experimental procedure consists in systematically adjusting the cell's inclination to achieve different incident angles, as well as changing the lamp's height to simulate different irradiance values. The resulting data will be analysed to quantify the impact of physical alignment on the cell's efficiency and fill factor.

Before executing the experimental procedure, a theoretical analysis was conducted to establish a baseline for the expected results. Based on the principles of optical absorption, the cell's behavior depends primarily on irradiance and inclination. The irradiance, G_{ph} , determines the available photon flux that reaches the semiconductor, through the equation (1).

$$G_{ph} = h \times f \times \phi(x) \quad (1)$$

An increase in irradiance, achieved by reducing the lamp's height, generates more electron-hole pairs, which should increase the short-circuit current, I_{sc} . Additionally, the cell's tilt angle affects the interface reflectance (R). Since the quantum efficiency is proportional to the term $(1 - R)$, equation (2), a larger inclination angle will increase reflection losses. Consequently, it is expected that as the tilt angle increases, less light will penetrate the cell, leading to a decrease in the cell's current and power output.

$$\eta = (1 - R) (1 - e^{-\alpha W}) \quad (2)$$

2 Results

The experimental setup used in this laboratory remained consistent with the methodology described in the previous laboratory experiments. The I-V curves were captured by performing a load sweep using a variable resistor connected in series to transition the cell's operating point across the curve.

This experiment was conducted for three different lamp heights, representing three different irradiance values (1026 W/m^2 , 603 W/m^2 and 320 W/m^2). For each irradiance value, the cell's I-V curves were captured for different tilt angles (0° , 20° , 40° and 65°).

After conducting the sweeps across different angles and heights, the oscilloscope data was imported into MATLAB, where a smooth trend line was applied to generate the I-V and P-V curves.

2.1 Tilting by the horizontal axis

Irradiation of 1026 W/m^2

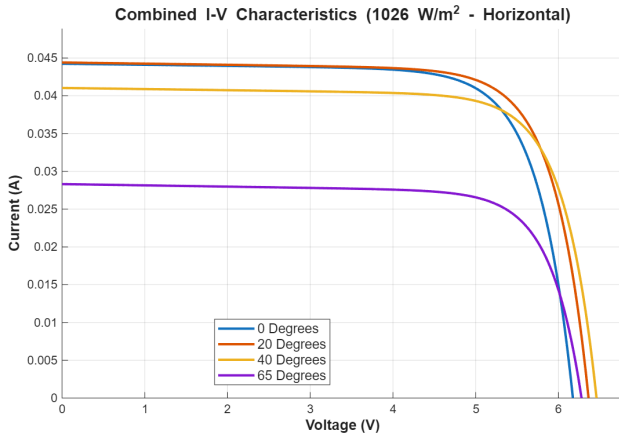


Figure 1: Combined I-V characteristics (1026 W/m^2 - Horizontal)

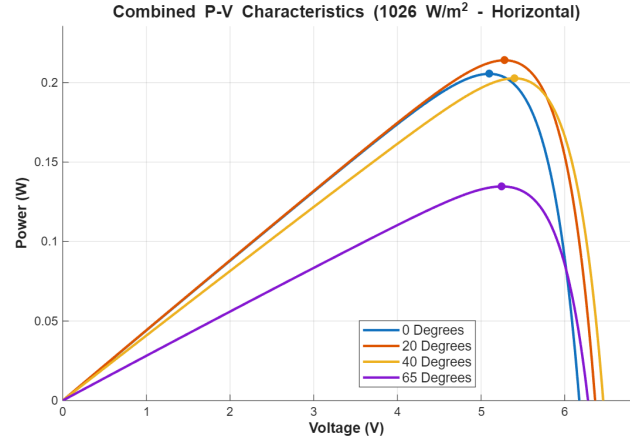


Figure 2: Combined P-V characteristics (1026 W/m^2 - Horizontal)

Table 1: Electrical parameters at 1026 W/m^2 (Horizontal)

Angle	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0°	6.1738	0.0442	0.2055	5.0974	0.0403	75.23	5.20
20°	6.3631	0.0444	0.2140	5.2792	0.0405	75.74	5.42
40°	6.4595	0.0410	0.2027	5.3980	0.0375	76.47	5.13
65°	6.2782	0.0283	0.1347	5.2465	0.0257	75.75	3.41

Irradiation of 603 W/m^2

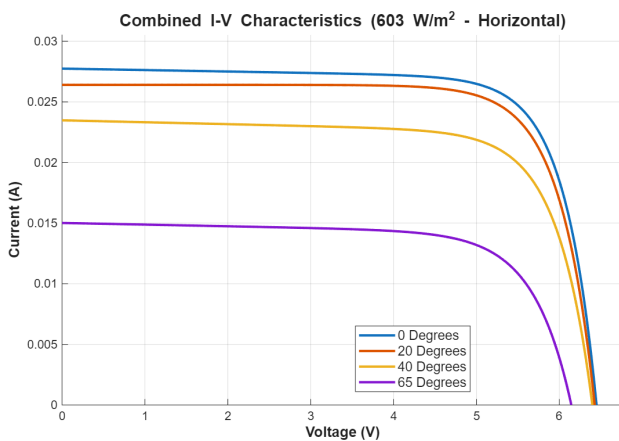


Figure 3: Combined I-V characteristics (603 W/m^2 - Horizontal)

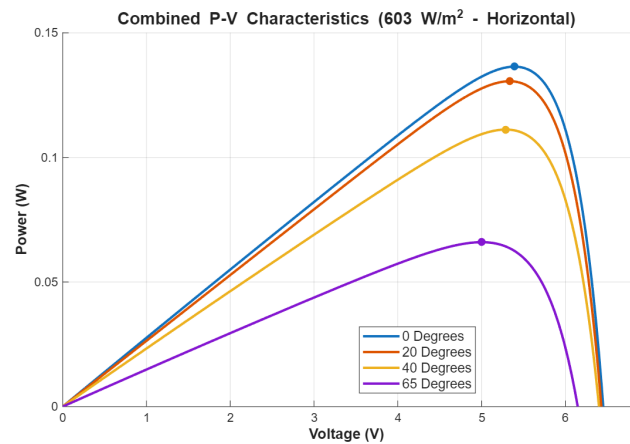
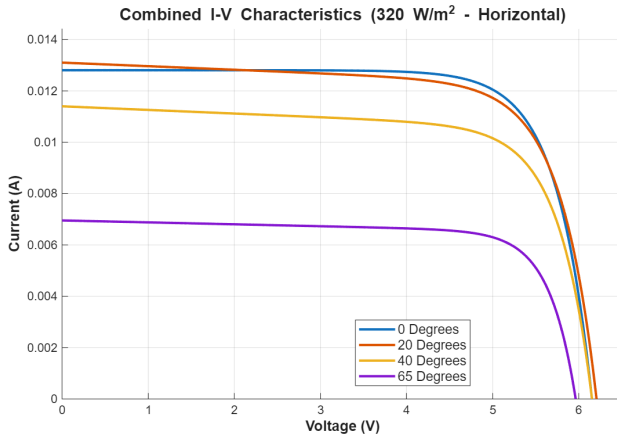
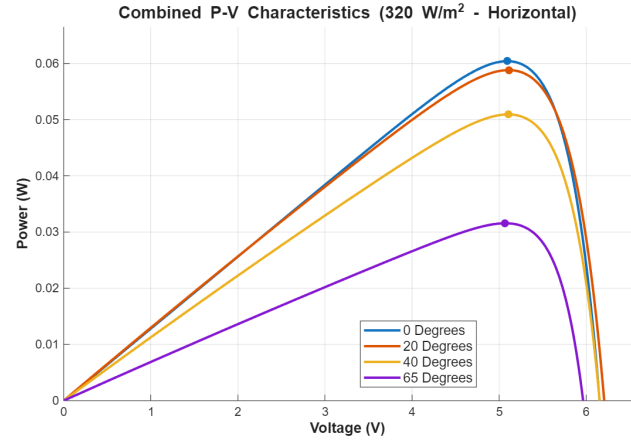


Figure 4: Combined P-V characteristics (603 W/m^2 - Horizontal)

Table 2: Electrical parameters at 603 W/m² (Horizontal)

Angle	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0°	6.4507	0.0277	0.1365	5.3906	0.0253	76.25	5.88
20°	6.4291	0.0264	0.1306	5.3339	0.0245	76.93	5.63
40°	6.4006	0.0235	0.1112	5.2847	0.0210	74.00	4.79
65°	6.1464	0.0150	0.0660	5.0009	0.0132	71.52	2.84

Irradiation of 320 W/m²**Figure 5:** Combined I-V characteristics (320 W/m² - Horizontal)**Figure 6:** Combined P-V characteristics (320 W/m² - Horizontal)**Table 3:** Electrical parameters at 320 W/m² (Horizontal)

Angle	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0°	6.1529	0.0128	0.0604	5.0924	0.0119	76.64	4.90
20°	6.2063	0.0131	0.0588	5.1118	0.0115	72.31	4.77
40°	6.1539	0.0114	0.0509	5.1057	0.0100	72.57	4.13
65°	5.9640	0.0070	0.0316	5.0676	0.0062	76.08	2.56

By analysing the graphs and the respective tables, it is possible to verify that both the irradiance and the tilt angle affect the current produced by the cell, and therefore the overall output power.

Regarding the variation in the lamp height, the expected trend is confirmed. As the irradiance decreases from 1026 W/m² to 603 W/m² and 320 W/m², both I_{sc} and P_{max} decrease significantly for all tilt angles. This happens because a lower irradiance leads to a lower photon flux reaching the photovoltaic cell, reducing the number of generated electron-hole pairs.

Regarding the tilt angle, it is possible to verify that increasing the inclination generally reduces the current and the maximum power, especially for 65°. This is explained by the increase in reflection losses and by the reduction of the effective irradiance absorbed by the cell.

For 1026 W/m², the best result is obtained at 20°, with $P_{max} = 0.2140$ W, while for 603 W/m² and 320 W/m² the highest values are obtained at 0°, with $P_{max} = 0.1365$ W and 0.0604 W, respectively. The fact that 20° gives a slightly higher power than 0° at the highest irradiance can be attributed

to experimental factors such as a small misalignment of the setup or a non-uniform light distribution from the lamp.

In general, the fill factor remains relatively stable for the lower tilt angles, while the efficiency decreases as the angle increases, reaching its lowest values at 65° . Therefore, the horizontal-axis results confirm that both lower irradiance and higher tilt angles reduce the electrical performance of the photovoltaic cell.

2.2 Tilting by the vertical axis

Irradiation of 1026 W/m^2

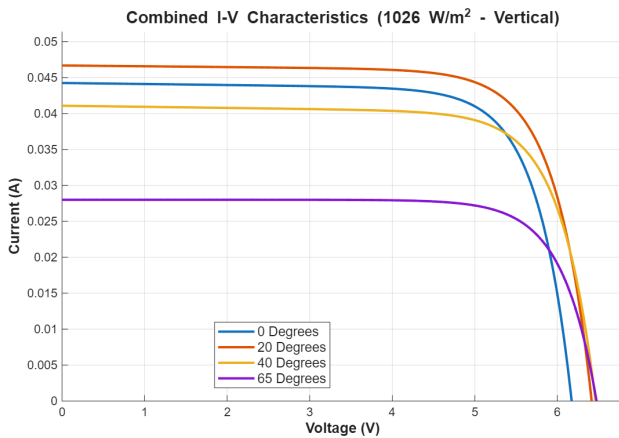


Figure 7: Combined I-V characteristics (1026 W/m^2 - Vertical)

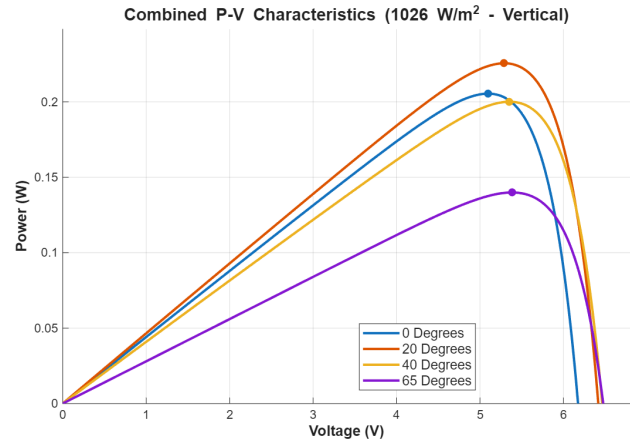


Figure 8: Combined P-V characteristics (1026 W/m^2 - Vertical)

Table 4: Electrical parameters at 1026 W/m^2 (Vertical)

Angle	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0°	6.1738	0.0442	0.2055	5.0974	0.0403	75.23	5.20
20°	6.4164	0.0467	0.2256	5.2848	0.0427	75.33	5.71
40°	6.4633	0.0411	0.2001	5.3494	0.0374	75.37	5.07
65°	6.4753	0.0280	0.1399	5.3853	0.0260	77.19	3.54

Irradiation of 603 W/m^2

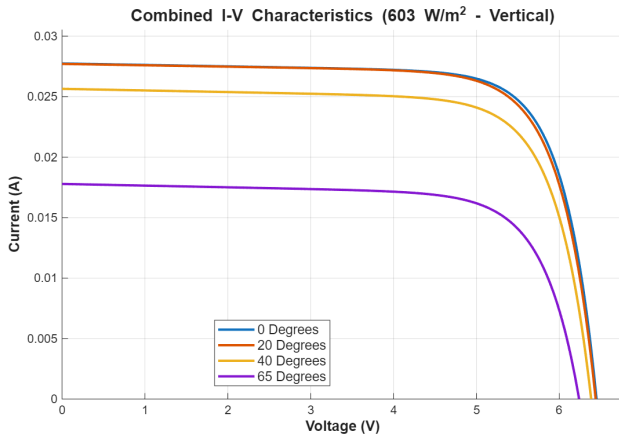


Figure 9: Combined I-V characteristics (603 W/m^2 - Vertical)

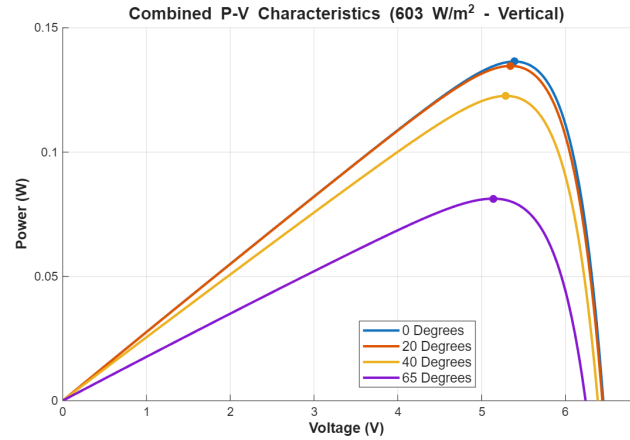


Figure 10: Combined P-V characteristics (603 W/m^2 - Vertical)

Table 5: Electrical parameters at 603 W/m^2 (Vertical)

Angle	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0°	6.4507	0.0277	0.1365	5.3906	0.0253	76.25	5.88
20°	6.4407	0.0277	0.1347	5.3436	0.0252	75.47	5.80
40°	6.3858	0.0256	0.1226	5.2853	0.0232	74.87	5.28
65°	6.2400	0.0178	0.0813	5.1396	0.0158	73.24	3.50

Irradiation of 320 W/m^2

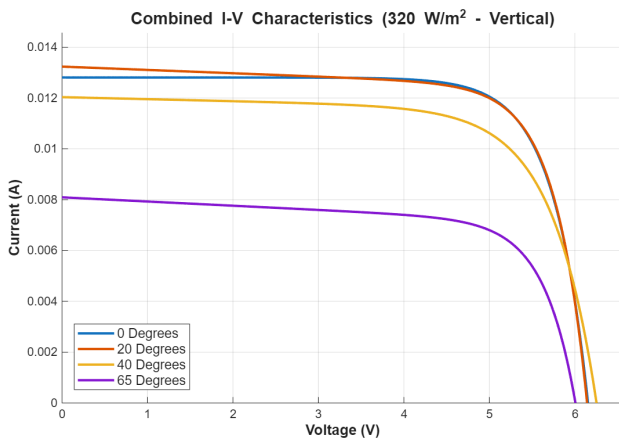


Figure 11: Combined I-V characteristics (320 W/m^2 - Vertical)

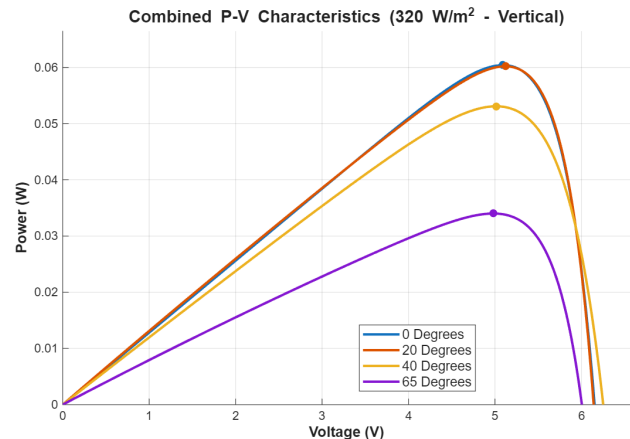


Figure 12: Combined P-V characteristics (320 W/m^2 - Vertical)

Table 6: Electrical parameters at 320 W/m² (Vertical)

Angle	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	FF (%)	Eff (%)
0°	6.1529	0.0128	0.0604	5.0924	0.0119	76.64	4.90
20°	6.1427	0.0132	0.0602	5.1210	0.0118	74.03	4.89
40°	6.2528	0.0120	0.0531	5.0123	0.0106	70.55	4.31
65°	6.0060	0.0081	0.0340	4.9829	0.0068	69.97	2.76

By analysing the results for the vertical-axis tilt, it is possible to observe the same general behaviour verified in the previous section. As the irradiance decreases, both I_{sc} and P_{max} also decrease for all tilt angles.

The effect of the tilt angle is also evident. Higher inclination angles, especially 65°, lead to lower current, power and efficiency. This behaviour is consistent with the increase in reflection losses and with the reduction of the effective radiation absorbed by the cell.

At 1026 W/m², the highest value of P_{max} is obtained at 20°, with 0.2256 W. At 603 W/m² and 320 W/m², the best results are obtained at 0°, with $P_{max} = 0.1365$ W and 0.0604 W, respectively. As in the horizontal-axis case, the slight improvement at 20° for the highest irradiance can be attributed to experimental factors such as setup alignment and lamp non-uniformity.

Overall, the vertical-axis configuration presents slightly better results than the horizontal-axis configuration, particularly for 1026 W/m² and at higher tilt angles.

2.3 Comparison

In order to better compare all the irradiance levels, tilt angles and axis configurations, the following table was performed.

Table 7: Electrical parameters for different irradiance levels, orientations and angles

Irr. (W/m ²)	Axis	Angle (°)	V_{oc} (V)	I_{sc} (A)	P_{max} (W)	V_{mp} (V)	I_{mp} (A)	Eff (%)
1026	H	0	6.1738	0.0442	0.2055	5.0974	0.0403	5.20
		20	6.3631	0.0444	0.2140	5.2792	0.0405	5.42
		40	6.4595	0.0410	0.2027	5.3980	0.0375	5.13
		65	6.2782	0.0283	0.1347	5.2465	0.0257	3.41
	V	0	6.1738	0.0442	0.2055	5.0974	0.0403	5.20
		20	6.4164	0.0467	0.2256	5.2848	0.0427	5.71
		40	6.4633	0.0411	0.2001	5.3494	0.0374	5.07
		65	6.4753	0.0280	0.1399	5.3853	0.0260	3.54
603	H	0	6.4507	0.0277	0.1365	5.3906	0.0253	5.88
		20	6.4291	0.0264	0.1306	5.3339	0.0245	5.63
		40	6.4006	0.0235	0.1112	5.2847	0.0210	4.79
		65	6.1464	0.0150	0.0660	5.0009	0.0132	2.84
	V	0	6.4507	0.0277	0.1365	5.3906	0.0253	5.88
		20	6.4407	0.0277	0.1347	5.3436	0.0252	5.80
		40	6.3858	0.0256	0.1226	5.2853	0.0232	5.28
		65	6.2400	0.0178	0.0813	5.1396	0.0158	3.50
320	H	0	6.1529	0.0128	0.0604	5.0924	0.0119	4.90
		20	6.2063	0.0131	0.0588	5.1118	0.0115	4.77
		40	6.1539	0.0114	0.0509	5.1057	0.0100	4.13
		65	5.9640	0.0070	0.0316	5.0676	0.0062	2.56
	V	0	6.1529	0.0128	0.0604	5.0924	0.0119	4.90
		20	6.1427	0.0132	0.0602	5.1210	0.0118	4.89
		40	6.2528	0.0120	0.0531	5.0123	0.0106	4.31
		65	6.0060	0.0081	0.0340	4.9829	0.0068	2.76

The evolution of the parameters V_{oc} , I_{sc} and P_{max} with the increasing of the tilting angle was analyzed with the figures presented below.

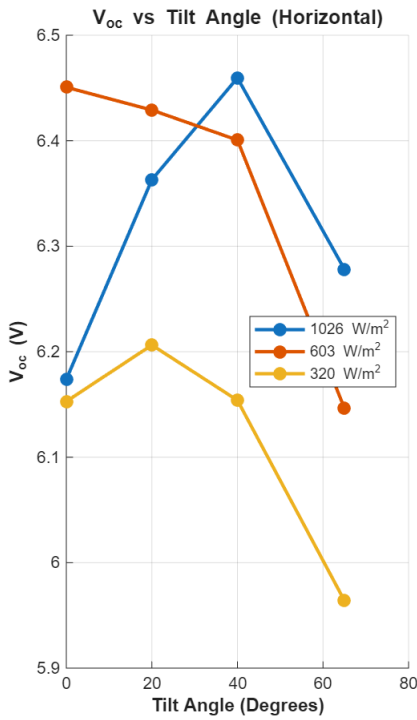


Figure 13: V_{oc} vs Tilt angle (Horizontal)

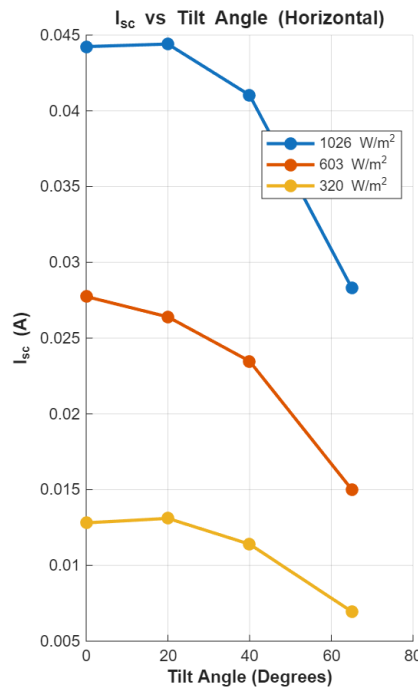


Figure 14: I_{sc} vs Tilt angle (Horizontal)

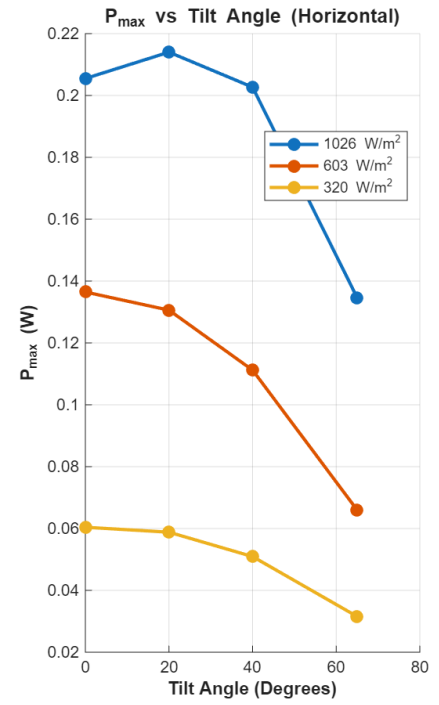


Figure 15: P_{max} vs Tilt angle (Horizontal)

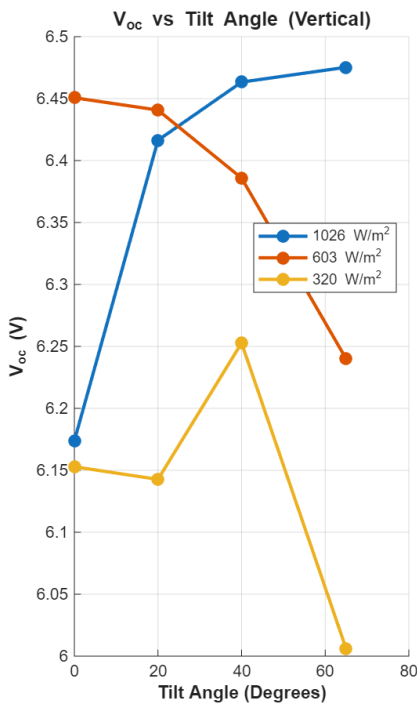


Figure 16: V_{oc} vs Tilt angle (Vertical)

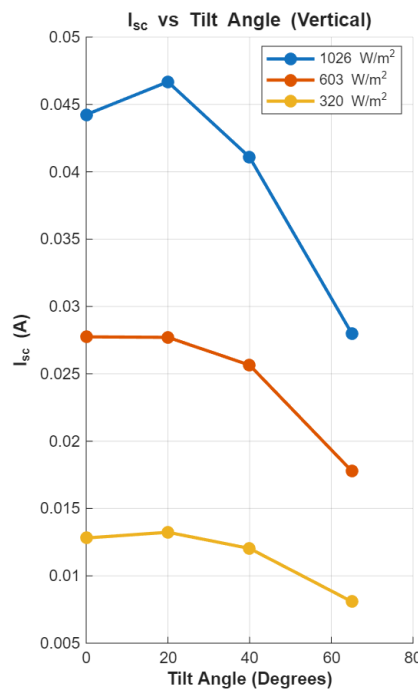


Figure 17: I_{sc} vs Tilt angle (Vertical)

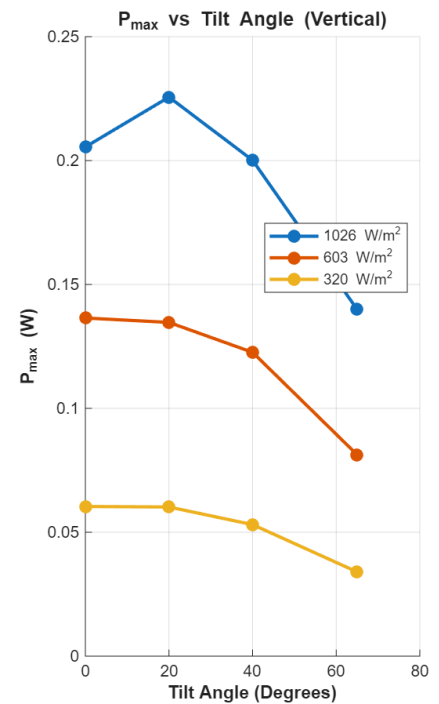


Figure 18: P_{max} vs Tilt angle (Vertical)

The comparison between the horizontal and vertical tilt configurations confirms the trends observed in the previous sections. For all irradiance levels, increasing the tilt angle generally leads to lower

values of I_{sc} , P_{max} and efficiency, with the largest degradation occurring at 65° .

For 1026 W/m^2 , the vertical-axis configuration presents the best overall result, with $P_{max} = 0.2256 \text{ W}$ at 20° , while the horizontal-axis configuration reaches a maximum of 0.2140 W at the same angle. For 603 W/m^2 , both configurations present very similar results at low angles, although the vertical-axis configuration performs better at higher inclinations. For 320 W/m^2 , the difference between both configurations is smaller, but the vertical-axis case still presents slightly better values at 40° and 65° .

These results suggest that the vertical-axis tilt configuration is, under the tested conditions, slightly more favourable than the horizontal-axis configuration. This can be explained by the geometry of the lamp-cell arrangement, which may produce lower optical losses for vertical tilting. Nevertheless, both configurations follow the expected theoretical behaviour: lower irradiance and higher tilt angles reduce the current generation capability of the cell and, consequently, the maximum power output.

Regarding efficiency, the same trend is observed. The highest values occur for low tilt angles and higher irradiance, while the lowest efficiencies are obtained at 65° . The fill factor does not vary as strongly as the current and power, remaining relatively stable in most cases, which indicates that the main effect of tilt angle is the reduction of absorbed irradiance rather than a strong distortion of the $I(V)$ curve shape.

2.4 Simulation of Energy produced in one day considering different tilt angles and irradiance levels

For this section, in order to compute the total energy produced in one day considering different angles and irradiance values, a dynamic sun model was created to emulate the sun's trajectory. Two scenarios were created: one for **summer**, with 14 hours of daylight (06:00 to 20:00), with a peak irradiance of 1000 W/m^2 and a minimum solar incidence angle of 15° at solar noon; one for **winter** with 9 hours of daylight (08:00 to 17:00), with a peak irradiance of 600 W/m^2 and a minimum solar incidence angle of 60° at solar noon.

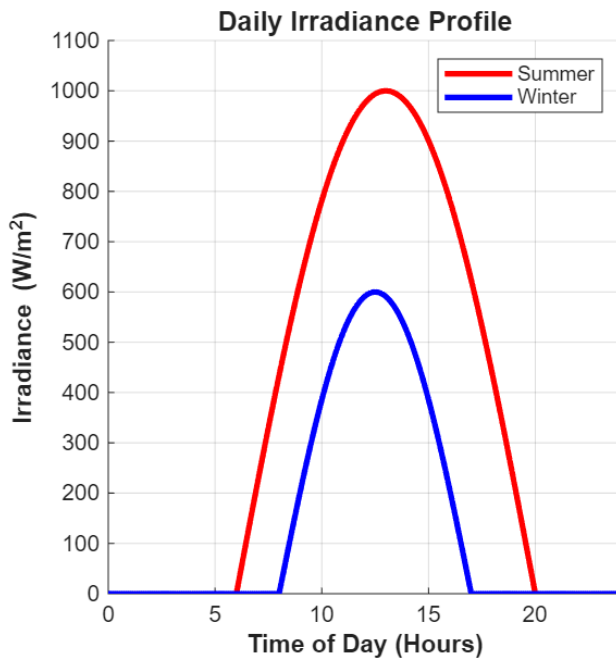


Figure 19: Daily irradiance profile

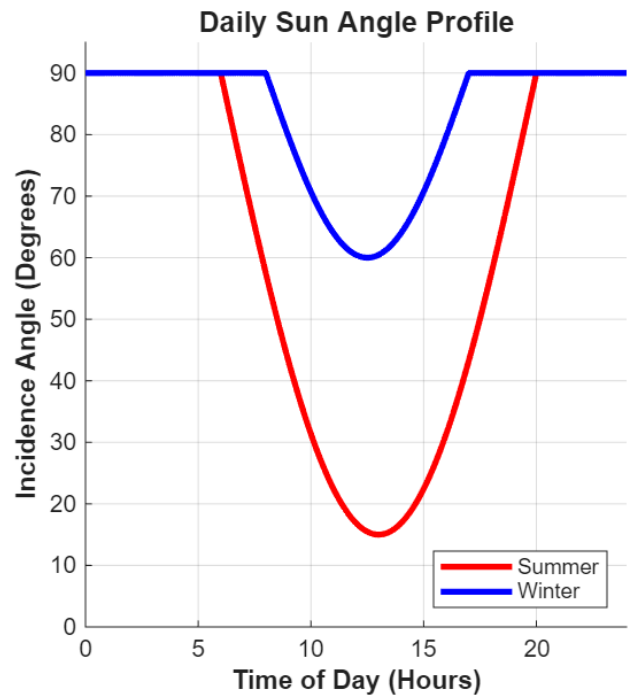


Figure 20: Daily sun angle profile

In real life the sun's angle and irradiance change simultaneously over time but the data only had specific points of tilt angle and irradiance, so the laboratory data was used to create a 3D interpolation model in MATLAB. With this model, maximum power (P_{max}), was estimated for any combination of irradiance and angle at any given hour of the simulated days.

The results obtained were the following.

Table 8: Total energy produced per unit area for dynamic sun conditions

Axis Configuration	Summer Energy (Wh/m ²)	Winter Energy (Wh/m ²)
Horizontal	435.89	91.41
Vertical	463.51	107.54

3 Conclusion

This laboratory experiment successfully evaluated the impact of irradiance levels, tilt angle and orientation on the performance of a photovoltaic cell.

As predicted, a reduction in the irradiance led to a decrease in short-circuit current, I_{sc} , and maximum power, P_{max} . This confirms that lower photon flux generates fewer electron-hole pairs.

Increasing the tilt angle, reduced the cell's performance, with the most significant power and efficiency drops occurring at 65°. This behavior validates the theoretical expectations that higher incident angles increase reflectance losses, reducing the effective irradiance that reaches the semiconductor.

Comparing the two orientations, the vertical-axis configuration presented slightly superior performance over the horizontal-axis configuration, particularly under high irradiance.

Finally, the dynamic 3D interpolation model provided real-world context to the laboratory data. The simulation highlighted the discrepancy in energy yield between summer and winter. The vertical-axis configuration yielded a higher total energy production in both seasons compared to the horizontal-axis configuration.

With this laboratory experiment it is important to retain that angle and orientation optimization is essential to maximize the output power of a photovoltaic system.